Bipolar Junction Transistors (BJTs)

Electronic Devices and Circuits

Comprehensive Study Report

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1 Introduction to Bipolar Junction Transistors

Before the invention of the transistor, electronic circuits relied on vacuum tubes, which were characteristically large, hot, and power-hungry, wasting a considerable amount of energy as heat. The landscape of electronics was revolutionized in 1947 when John Bardeen, Walter Brattain, and William Shockley invented the transistor at Bell Laboratories.

While the Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) became familiar in the 1970s and 1980s and is currently the most widely used device in integrated circuit (IC) design (especially via CMOS technology), the Bipolar Junction Transistor (BJT) remains the preferred device in highly demanding applications. BJTs are particularly essential in high-frequency analog and high-speed digital integrated circuits.

Furthermore, BJTs and MOSFETs can be combined to create an innovative technology known as **BiCMOS**. This hybrid technology takes advantage of the high input impedance and low-power operation of CMOS, paired with the very high-frequency response and superior current-driving capability of the BJT.

2 Device Structure and Physical Operation

2.1 Physical Structure

The BJT consists of three doped semiconductor regions: the **Emitter (E)**, the **Base (B)**, and the **Collector (C)**.

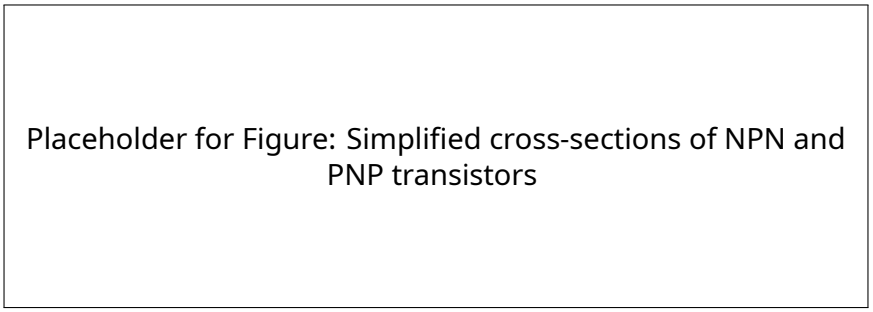


Figure 1: Simplified structure and schematic symbols of NPN and PNP transistors.

These three regions form two pn-junctions:

1. **Emitter-Base Junction (EBJ):** Between the emitter and the base.
2. **Collector-Base Junction (CBJ):** Between the collector and the base.

BJTs are fabricated in two polarities: **npn** and **pnp**. In an *npn* transistor, a thin p-type base region is sandwiched between an n-type emitter and an n-type collector. The device is asymmetric by design: the emitter is heavily doped (to inject a large number of charge carriers) and the base is very lightly doped and extremely thin (typically in the nanometer range in modern ICs). Because charge carriers of both polarities (electrons and holes) participate in the conduction process, the device is termed a *bipolar* transistor.

2.2 Modes of Operation

Depending on the bias (forward or reverse) applied to the EBJ and CBJ, a BJT operates in one of several modes:

Mode	Emitter-Base Junction (EBJ)	Collector-Base Junction (CBJ)
Active	Forward	Reverse
Saturation	Forward	Forward
Cut-off	Reverse	Reverse

Table 1: BJT Modes of Operation

The **active mode** is used when the transistor is required to operate as a linear amplifier. **Cut-off** and **saturation modes** are primarily used for switching applications, such as in digital logic circuits.

2.3 Operation in the Forward-Active Mode

Consider an npn transistor biased in the active mode ($V_{BE} > 0$, $V_{CB} \geq 0$). The forward-biased EBJ causes electrons to be injected from the heavily doped n-type emitter into the p-type base. Simultaneously, holes are injected from the base into the emitter.

Because the base is exceedingly thin, most of the injected electrons do not recombine with holes. Instead, they diffuse across the base and are swept into the collector by the reverse-biased CBJ's electric field.

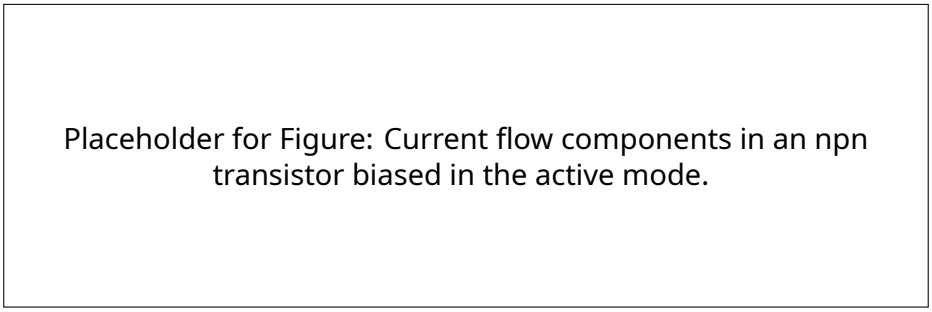


Figure 2: Carrier injection, diffusion, and collection in the active mode.

2.3.1 Minority-Carrier Distribution

The electron concentration in the p-type base at the edge of the EBJ depletion region, $n_p(0)$, is proportional to e^{V_{BE}/V_T}, where V_T is the thermal voltage. Because the base is very thin, this concentration decays almost linearly to zero at the edge of the CBJ. The electron diffusion current I_n is proportional to the slope of this profile.

2.3.2 Current Relationships

Collector Current (i_C): The electrons that successfully diffuse across the base and reach the collector form the collector current:

$$i_C = I_S e^{V_{BE}/V_T}$$

where I_S is the saturation current (or scale current), which is directly proportional to the cross-sectional area of the EBJ and inversely proportional to the base width W.

Base Current (i_B): The base current consists of two components: holes injected from the base into the emitter (i_{B1}), and holes supplied by the external circuit to replace those

lost to recombination in the base (i_{B2}). Both components are proportional to e^{V_{BE}/V_T}. The total base current is expressed as a fraction of the collector current:

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta}\right) e^{V_{BE}/V_T}$$

Here, β is the **common-emitter current gain**. To achieve a high β (typically 50 to 200, though sometimes up to 1000), the base must be thin and lightly doped, and the emitter heavily doped.

Emitter Current (i_E): By Kirchhoff's Current Law, the emitter current is the sum of the collector and base currents:

$$i_E = i_C + i_B = i_C + \frac{i_C}{\beta} = \left(\frac{\beta+1}{\beta}\right) i_C$$

We can also define the **common-base current gain**, α:

$$i_C = \alpha i_E \quad \Longrightarrow \quad \alpha = \frac{\beta}{\beta+1}, \quad \beta = \frac{\alpha}{1-\alpha}$$

2.4 Large-Signal Equivalent Circuit Models

The physical operation of the BJT can be modeled using a controlled current source and a diode. The basic large-signal model consists of a diode D_E (representing the EBJ) between the base and emitter, and a controlled current source αi_E or βi_B between the collector and emitter.

Placeholder for Figure: Large-signal equivalent-circuit models of the npn BJT.

Figure 3: DC large-signal models for the active mode.

3 BJT Current–Voltage Characteristics

3.1 Base-Emitter Characteristics

The i_C versus V_{BE} characteristic is an exponential curve identical to a diode. A critical property of the EBJ is its temperature dependence: for a constant collector current, V_{BE} decreases by approximately 2 mV for every 1°C rise in temperature.

3.2 Collector Characteristics and the Early Effect

The i_C versus V_{CE} output characteristics reveal that when V_{CE} is sufficiently high to reverse-bias the CBJ ($V_{CE} > V_{BE} - 0.4$ V), the transistor operates in the active region. However, the collector current is not perfectly constant.

As V_{CE} increases, the reverse bias across the CBJ increases, which widens the CBJ depletion region. This effectively narrows the neutral base width W, increasing the concentration gradient and thus increasing i_C. This phenomenon is known as **base-width modulation** or the **Early Effect**.

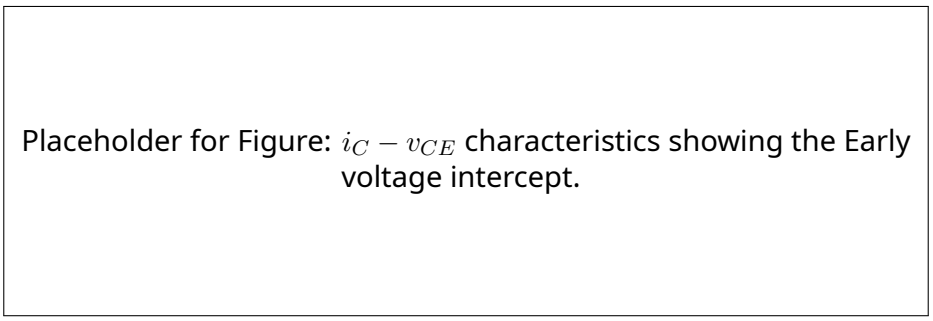


Figure 4: The Early effect and finite output resistance in a practical BJT.

Extrapolating the active-region characteristic lines backward reveals they intersect the negative V_{CE} axis at a common point, $-V_A$, where V_A is the **Early voltage** (typically 10 V to 100 V). The linear dependence of i_C on V_{CE} is modeled as:

$$i_C = I_S e^{V_{BE}/V_T}\left(1 + \frac{V_{CE}}{V_A}\right)$$

This finite slope introduces an output resistance, r_o, defined at a given bias point I_C:

$$r_o = \frac{V_A}{I_C}$$

3.3 Saturation Characteristics

If V_{CE} drops below a few tenths of a volt (specifically, when the CBJ becomes forward biased), the transistor enters saturation. The collector can no longer efficiently sweep out carriers. The incremental β drops significantly below its active-region value. A saturated BJT is often modeled simply with $V_{CE(sat)} \approx 0.2$ V and $V_{BE} \approx 0.7$ V.

4 DC Analysis of BJT Circuits

When analyzing DC circuits, we typically assume the transistor is in the active mode, calculate the terminal currents and voltages, and then verify the assumption by checking that $V_{CE} \geq 0.3$ V (for npn).

4.1 Worked Examples

Example 1: Temperature and Current Dependence of V_{BE} Consider a silicon npn transistor with $V_{BE} = 0.7$ V at $I_C = 1$ mA. We wish to find V_{BE} at $I_C = 0.1$ mA and 10 mA at room temperature ($V_T \approx 25$ mV). Using the exponential relationship:

$$\frac{I_{C2}}{I_{C1}} = e^{(V_{BE2} - V_{BE1})/V_T} \quad \Longrightarrow \quad V_{BE2} = V_{BE1} + V_T \ln\left(\frac{I_{C2}}{I_{C1}}\right)$$

For $I_{C2} = 0.1$ mA:

$$V_{BE2} = 0.7 + 0.025 \ln(0.1) = 0.7 - (0.025)(2.303) \approx 0.642 \text{ V}$$

For $I_{C2} = 10$ mA:

$$V_{BE2} = 0.7 + 0.025 \ln(10) = 0.7 + (0.025)(2.303) \approx 0.758 \text{ V}$$

5 Small-Signal Operation and Models

To use the BJT as a linear amplifier, it is biased at a DC quiescent operating point (Q-point) in the active region, and a small AC signal v_{be} is superimposed on V_{BE}.

5.1 Linearization and Transconductance

The total collector current is:

$$i_C = I_S e^{(V_{BE}+v_{be})/V_T} = I_S e^{V_{BE}/V_T} e^{v_{be}/V_T} = I_C e^{v_{be}/V_T}$$

If the signal is sufficiently small ($v_{be} \ll V_T$, usually limited to 5–10 mV), we can approximate the exponential via a Taylor series:

$$i_C \approx I_C \left(1 + \frac{v_{be}}{V_T}\right) = I_C + \frac{I_C}{V_T} v_{be} = I_C + i_c$$

The AC signal current i_c is linearly related to v_{be} by the **transconductance**, g_m:

$$g_m = \left.\frac{\partial i_C}{\partial v_{BE}}\right|_{I_C} = \frac{I_C}{V_T}$$

Graphically, g_m is the slope of the tangent to the $i_C - v_{BE}$ curve at the Q-point.

5.2 Input Resistances

The small-signal input resistance looking into the base is r_π:

$$r_\pi \equiv \frac{v_{be}}{i_b} = \frac{v_{be}}{i_c/\beta} = \frac{\beta}{g_m} = \frac{V_T}{I_B}$$

Alternatively, the resistance looking into the emitter is r_e:

$$r_e \equiv \frac{v_{be}}{i_e} = \frac{V_T}{I_E} = \frac{\alpha}{g_m} \approx \frac{1}{g_m}$$

These resistances are related by the fundamental property of the BJT:

$$r_\pi = (\beta + 1) r_e$$

5.3 AC Equivalent Circuits

To perform AC analysis, DC voltage sources are short-circuited to ground, and DC current sources are open-circuited. The transistor is replaced by a small-signal model.

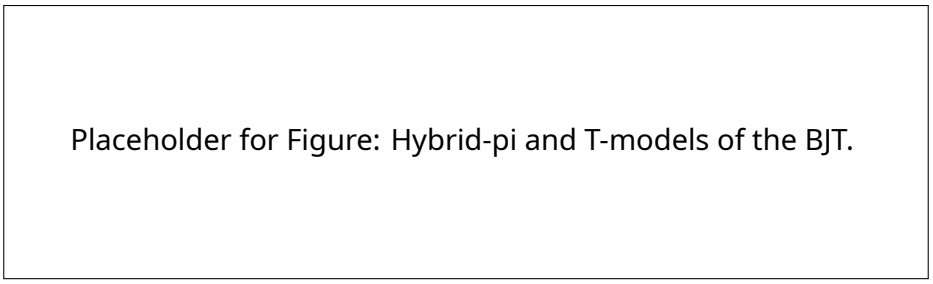


Figure 5: The Hybrid-π (left) and T-Model (right) small-signal equivalent circuits.

The **Hybrid-π model** uses r_π between base and emitter and a controlled source $g_m v_\pi$ (where $v_\pi = v_{be}$). The **T-model** uses r_e in the emitter leg. Both models can naturally incorporate the output resistance $r_o = V_A/I_C$ across the collector and emitter terminals.

6 Basic Amplifier Configurations

BJTs can be configured in three fundamental amplifier topologies: Common-Emitter (CE), Common-Base (CB), and Common-Collector (CC, or Emitter Follower).

To characterize an amplifier, we define:

- **Input Resistance (R_{in}):** The resistance looking into the amplifier, determining how much of the source signal v_{sig} actually reaches the input ($v_i = v_{sig}\frac{R_{in}}{R_{in}+R_{sig}}$).
- **Open-Circuit Voltage Gain (A_{vo}):** The gain v_o/v_i without a load.
- **Output Resistance (R_o):** The resistance looking back into the output, forming a divider with the load R_L.
- **Overall Voltage Gain (G_v):** The total gain from the signal source to the load:

$$G_v = \frac{v_o}{v_{sig}} = \left(\frac{R_{in}}{R_{in}+R_{sig}}\right) A_{vo} \left(\frac{R_L}{R_L+R_o}\right)$$

7 Common-Emitter Amplifiers

7.1 Basic Common-Emitter (CE) Amplifier

In the basic CE amplifier, the emitter is grounded.

- $R_{in} = r_\pi$
- $A_{vo} = -g_m R_C$
- $R_o = R_C$ (or $R_C \parallel r_o$ if the Early effect is included).
- Overall gain with load: $A_v = -g_m(R_C \parallel R_L)$

The CE amplifier provides high voltage gain but has a moderate input resistance and suffers from a relatively limited high-frequency response.

7.2 CE Amplifier with Emitter Degeneration Resistance (R_e)

Inserting a resistance R_e in the emitter leg introduces negative feedback (emitter degeneration). Analyzing this with the T-model yields the **resistance-reflection rule**: the input resistance looking into the base is $(\beta+1)$ times the total resistance in the emitter circuit.

$$R_{in} = (\beta+1)(r_e + R_e)$$

The voltage gain becomes:

$$A_{vo} = -\frac{\alpha R_C}{r_e + R_e} \approx -\frac{g_m R_C}{1 + g_m R_e}$$

Trade-offs: Including R_e decreases the voltage gain by a factor of $(1 + g_m R_e)$, but it increases the input resistance by the same factor. Furthermore, it vastly improves the linearity of the amplifier (allowing larger input signal swings) and makes the overall gain less sensitive to variations in β.

8 Transistor Breakdown and Temperature Effects

8.1 Breakdown Voltages

When a junction reverse bias becomes too large, avalanche breakdown occurs.

- BV_{CBO}: The breakdown voltage of the CBJ with the emitter open-circuited. Typically > 50 V.
- BV_{CEO}: The breakdown voltage across the collector and emitter with the base open. Because of transistor action, BV_{CEO} is significantly lower—about half the value of BV_{CBO}. This is sometimes called the *sustaining voltage*.

8.2 Current and Temperature Dependence of β

The current gain β is not a universal constant for a transistor.

- At very low collector currents, recombination in the EBJ space-charge region dominates, keeping β small.
- At moderate currents ($I_C \approx 1$ mA), β reaches its peak.
- At very high currents, high-level injection effects cause β to decrease.

Additionally, β increases with temperature. Because an increase in temperature causes V_{BE} to drop (for a constant current) and β to rise, I_C can increase significantly. This increased current generates more heat, potentially leading to thermal runaway. Proper bias stabilization is thus critical in BJT circuit design.

9 Formula Sheet

Concept / Parameter	Equation
Active-Mode DC Currents	$I_C = I_S e^{V_{BE}/V_T}$
	$I_B = \frac{I_C}{\beta}, \quad I_E = \frac{I_C}{\alpha}$
Alpha and Beta Relations	$\alpha = \frac{\beta}{\beta+1}, \quad \beta = \frac{\alpha}{1-\alpha}$
Thermal Voltage	$V_T = \frac{kT}{q} \approx 25$ mV (at room temp)
Early Effect	$I_C = I_S e^{V_{BE}/V_T}\left(1 + \frac{V_{CE}}{V_A}\right)$
Small-Signal Transconductance	$g_m = \frac{I_C}{V_T}$
Small-Signal Input Resistances	$r_\pi = \frac{\beta}{g_m} = \frac{V_T}{I_B}$
	$r_e = \frac{\alpha}{g_m} = \frac{V_T}{I_E}$
	$r_\pi = (\beta + 1) r_e$
Small-Signal Output Resistance	$r_o = \frac{V_A}{I_C}$
CE Voltage Gain (No R_e)	$A_v = -g_m(R_C \parallel R_L)$
CE Input Resistance (With R_e)	$R_{in} = (\beta + 1)(r_e + R_e)$
CE Voltage Gain (With R_e)	$A_{vo} \approx -\frac{g_m R_C}{1 + g_m R_e}$

Table 2: Key BJT Equations and Relationships